

# PIONEER JUNIOR COLLEGE

JC2 PRELIMINARY EXAMINATION  
HIGHER 2

CANDIDATE  
NAME

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CT  
GROUP

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INDEX  
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# CHEMISTRY

**9729/03**

### Paper 3 Free Response

**19 September 2018**

**2 hours**

Candidates answer on separate paper.

Additional Materials:      Answer Paper  
                                         Data Booklet  
                                         Cover Page

## READ THESE INSTRUCTIONS FIRST

Write your name, index number and CT group on all work you hand in.  
Write in dark blue or black pen on both sides of the paper.  
You may use a soft pencil for any diagrams, graphs or rough workings.  
Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A  
Answer **all** questions.

Section B  
Answer **one** question.

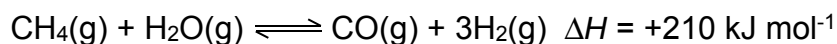
The use of an approved scientific calculator is expected, where appropriate. A Data Booklet is provided.

At the end of the examination, fasten all your work securely together.  
The number of marks is given in brackets [ ] at the end of each question or part question.

## Section A

Answer all questions in this section.

- 1 (a) Steam reforming is an important industrial method for producing hydrogen and carbon monoxide, from hydrocarbon fuels. This is achieved in a processing device called a reformer which reacts steam at high temperature with the fuel, such as methane gas.



- (i) Write an expression of  $K_p$  for the reaction, indicating its units. [2]
- (ii) At temperature  $T$ , a mixture of gaseous methane, steam and carbon monoxide were introduced into an evacuated reactor.

The table below shows the initial pressures of the gases before they were added to the reactor.

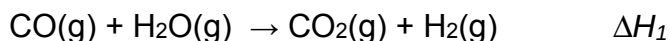
| Gas                          | Initial pressure / atm |
|------------------------------|------------------------|
| methane, $\text{CH}_4$       | 1.50                   |
| steam, $\text{H}_2\text{O}$  | 2.50                   |
| carbon monoxide, $\text{CO}$ | 0.70                   |

Given that the percentage dissociation of methane is 20% at equilibrium, find the  $K_p$  for the reaction. [2]

- (iii) A chemist decided to raise the temperature of the reactor. He suggested that this would result in a lower yield of the product.

Do you agree with the chemist? Explain your answer. [2]

In order to produce more hydrogen gas, the carbon monoxide gas that is produced from the steam reforming reaction can further react with steam. This is known as the “water-gas shift” reaction.



- (iv) Given that the reaction conditions were set at  $500^\circ\text{C}$  and 1.5 atm, determine the maximum volume of hydrogen gas that can be produced from 5kg of carbon monoxide gas.

State the assumption made in your calculation. [3]

(v) Given the following,

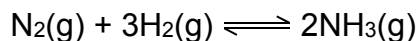
|                                                                            |      |
|----------------------------------------------------------------------------|------|
| enthalpy change of formation of CO(g) / kJ mol <sup>-1</sup>               | -99  |
| enthalpy change of formation of H <sub>2</sub> O(g) / kJ mol <sup>-1</sup> | -245 |
| enthalpy change of formation of CO <sub>2</sub> (g) / kJ mol <sup>-1</sup> | -394 |

With reference to the above data, calculate the enthalpy change of reaction of the “water-gas shift” reaction,  $\Delta H_1$ .

[1]

(b) The hydrogen gas obtained from the steam reforming process in part (a) can be reacted with nitrogen gas to produce ammonia gas.

This reaction is known as the Haber-Bosch Process, using iron as a catalyst.



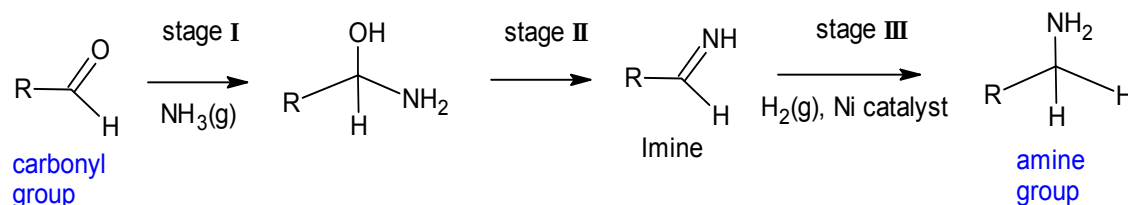
(i) Predict and explain the sign of  $\Delta S$  for the Haber-Bosch reaction. [1]

(ii) Using your answer in (b)(i), calculate the maximum temperature for the Haber-Bosch reaction to be spontaneous.

It is known that the numerical value of  $\Delta S$  is 198 J K<sup>-1</sup> mol<sup>-1</sup>, and that  $\Delta H = -92.0$  kJ mol<sup>-1</sup>. [2]

- (c) The hydrogen gas obtained from steam reforming, and the ammonia gas obtained from the Haber-Bosch reaction, are the key reagents required for the reductive amination reaction. Many amines in the pharmaceutical drugs are synthesised through this reaction.

The diagram, **Fig. 1.1**, shows the 3 stages involved in reductive amination.



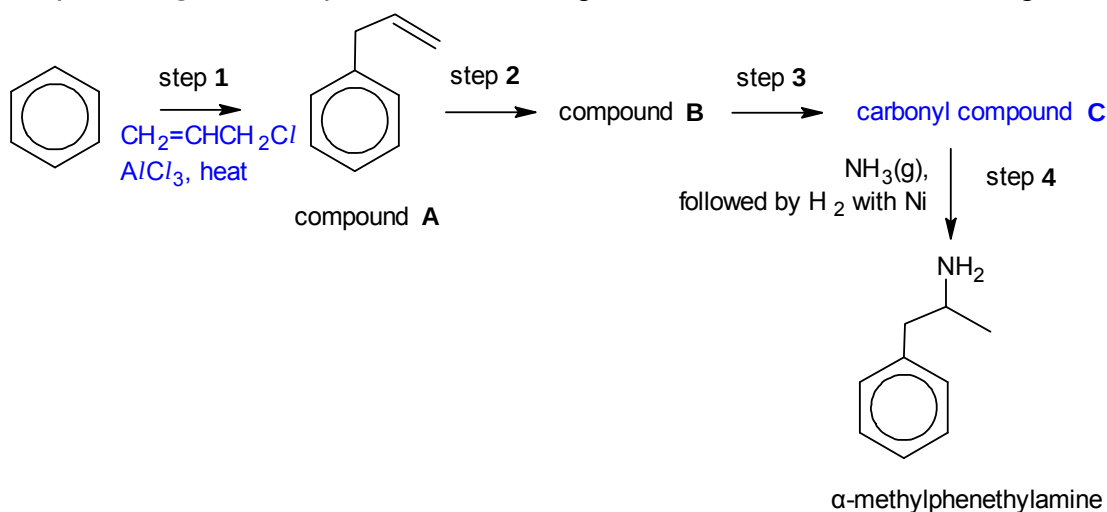
**Fig. 1.1**

During the reaction, a carbonyl group is converted into an amine group via an intermediate imine.

- (i) State the types of reaction occurring in stage I and stage II. [2]

The reaction scheme below, **Fig.1.2**, shows the production of  $\alpha$ -methylphenethylamine, which involves reductive amination in step 4.

Step 4 in **Fig. 1.2** comprises of the 3 stages of reductive amination in Fig. 1.1.



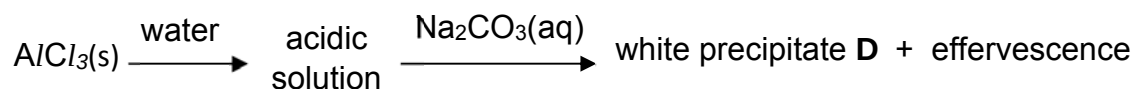
**Fig.1.2**

- (ii) In step 1, benzene undergoes a reaction with 3-chloroprop-1-ene with the aid of anhydrous aluminium chloride catalyst. Name and describe the mechanism for step 1, showing clearly any relevant charges, dipoles or lone pairs of electrons that are important. [4]
- (iii) Deduce and draw the structure for carbonyl compound C, and suggest the structure of compound B. [2]

[Total: 21]

2 This question is on halogen-containing compounds.

(a) The following experiments are carried out starting from solid aluminium chloride.



(i) Explain with the aid of balanced equations, why  $\text{AlCl}_3(\text{s})$  dissolves to form an acidic solution. Suggest the pH of the acidic solution. [3]

(ii) Identify precipitate **D**. [1]

(b) When aqueous  $\text{AgNO}_3$  is added to separate test-tubes containing aqueous  $\text{KCl}$  and  $\text{KI}$ , the various  $\text{AgX}$  ( $\text{X} = \text{Cl}, \text{I}$ ) precipitates were observed.

With the aid of relevant equations, explain why a precipitate of  $\text{AgCl}$  is soluble in excess aqueous  $\text{NH}_3$  but a precipitate of  $\text{AgI}$  is insoluble in excess aqueous  $\text{NH}_3$ . [3]

(c) The standard redox potential of chlorate ( $\text{V}$ ) ions is given below.



Predict what you expect to observe, when acidified potassium chlorate ( $\text{V}$ ) is added separately to each of the following reagents.

State clearly if no reaction occurs. Write a balanced equation for any reaction that occurs. Explain your answer using  $E^\ominus$  values from the *Data Booklet*.

(i)  $\text{FeSO}_4 (\text{aq})$

(ii) Acidified  $\text{KMnO}_4 (\text{aq})$

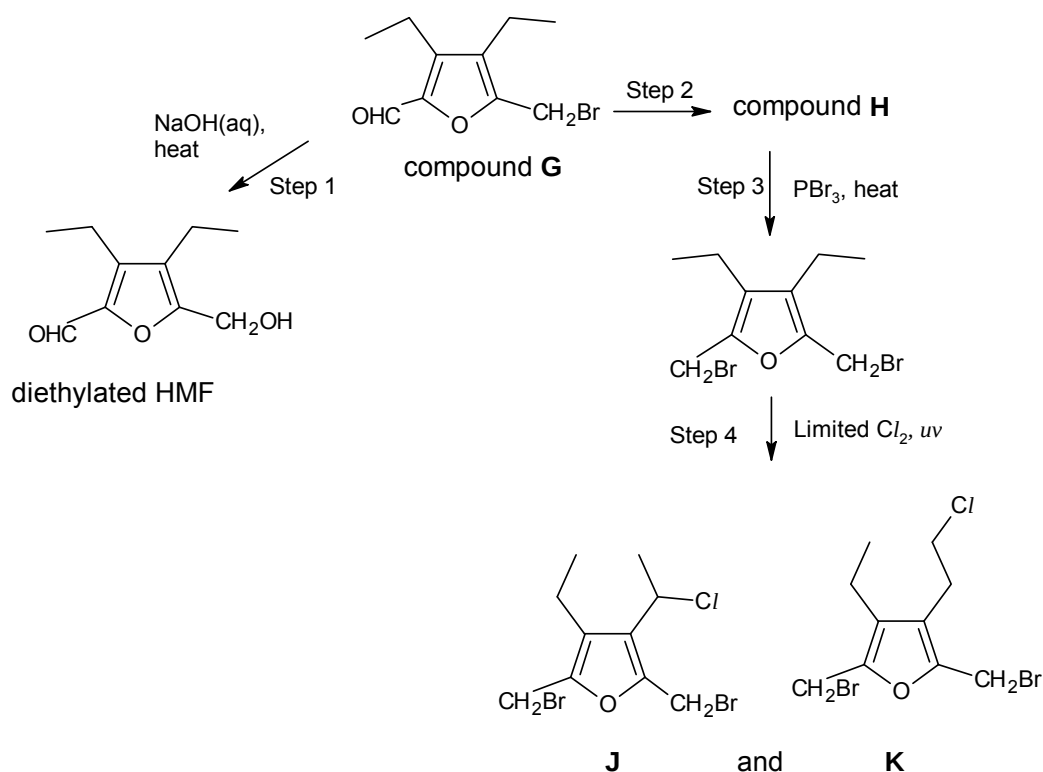
[3]

(d)  $\text{PCl}_5$  reacts with hydrazine,  $\text{N}_2\text{H}_4$ , to give compound **E**, which has the following composition by mass: P, 20.5%; N, 9.2%; Cl, 70.3%. The molecule of **E** contains an N-N single bond.

Calculate the empirical formula of **E**, and suggest its molecular formula and the structure of its molecule, assuming its constituent atoms show their usual valencies.

[2]

- (e) Diethylated HMF is used to synthesize a variety of pharmaceuticals. The flow-scheme below shows the reactions involving diethylated HMF and compound **G**, its precursor. Assume the C-O-C ether group is inert.



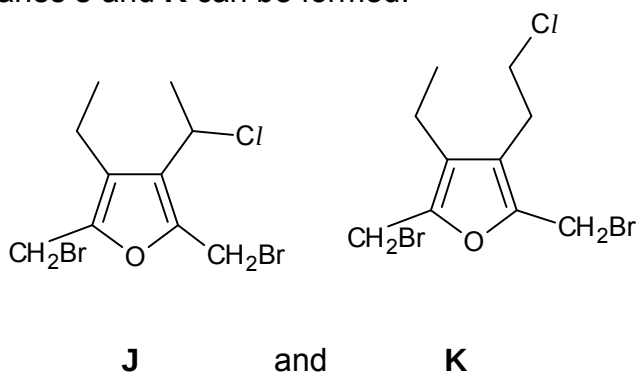
- (i) When the reaction in step **1** was investigated kinetically, it was found that its rate was independent of [OH<sup>-</sup>], but was first order with respect to [compound **G**].

Name and describe a mechanism for this reaction, showing curly arrows in your answer. [3]

You may represent compound **G** as R-CH<sub>2</sub>Br.

- (ii) Draw the structure of compound **H**. [1]

- (iii) When the mono-chlorination is carried out in step 4, two possible monochloroalkanes **J** and **K** can be formed.

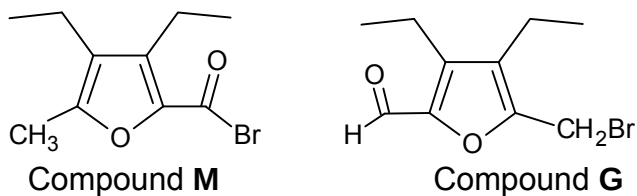


Predict the relative proportions of **J** and **K** formed. [1]

- (iv) When the monochlorinated products of step 4 were analysed, it is found that the mole ratio of **J**: **K** formed was about 97: 3.

Suggest a reason for the difference between this ratio and the one you gave in (iii ). [1]

- (v) Compounds **M** and **G** are both bromine-containing compounds that are isomers of each other.



Compare the reactivities of **M** and **G** with water. Explain your answer. [2]  
[Total:20]

- 3 (a) (i) Use the following data to construct an energy level diagram and hence calculate the enthalpy change for the hydrogenation of ethene gas to ethane gas.

|                                                                | $\Delta H / \text{kJ mol}^{-1}$ |
|----------------------------------------------------------------|---------------------------------|
| Enthalpy change of combustion of $\text{CH}_3\text{CH}_3(l)$   | -1545                           |
| Enthalpy change of condensation of $\text{CH}_3\text{CH}_3(g)$ | -15                             |
| Enthalpy change of combustion of $\text{H}_2(g)$               | -286                            |
| Enthalpy change of combustion of $\text{CH}_2=\text{CH}_2(g)$  | -1411                           |

[4]

- (ii) Define the term bond energy. [1]

- (iii) Using appropriate bond energy values from the *Data Booklet* alone, calculate another value for the enthalpy change of the reaction in (a)(i). [2]

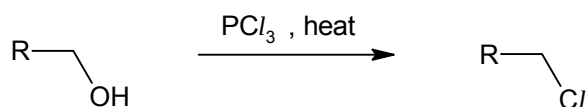
- (iv) Suggest a reason for the discrepancy between the enthalpy change for the hydrogenation of ethene gas to ethane gas calculated in (a)(i) and (a)(iii). [1]

- (b) Hot sulfuric acid,  $\text{H}_2\text{SO}_4$  is added to ethanol, an alcohol, to produce ethene.

Draw a 'dot-and-cross' diagram showing the electrons (outer shell only) in a  $\text{H}_2\text{SO}_4$  molecule.

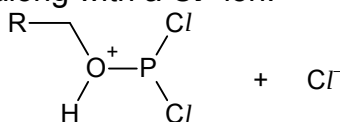
Use the VSEPR (valence shell electron pair repulsion) theory to predict the bond angle around sulfur in  $\text{H}_2\text{SO}_4$ . [3]

- (c) Phosphorous halides such as  $\text{PX}_3$  can act as both Lewis acids or Lewis bases.  $\text{PCl}_3$  is used to convert an alcohol to an alkyl chloride.



The mechanism above involves 2 steps:

- Step 1: The nucleophilic alcohol attacks  $\text{PCl}_3$  to form the following intermediate, along with a  $\text{Cl}^-$  ion:



- Step 2: (slow)  $\text{Cl}^-$  ion attacks the electrophilic carbon on the intermediate, simultaneously the C–O bond breaks heterolytically.

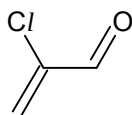
State and explain if  $\text{PCl}_3$  acts as a Lewis acid or Lewis base in step 1 of the mechanism. [1]



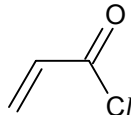
- (d) For the following pair of compounds, describe **one** chemical test which would enable you to distinguish clearly between them.

State the observations for **each** compound.

State the type of reaction undergone for the compound that shows a positive test.



compound N

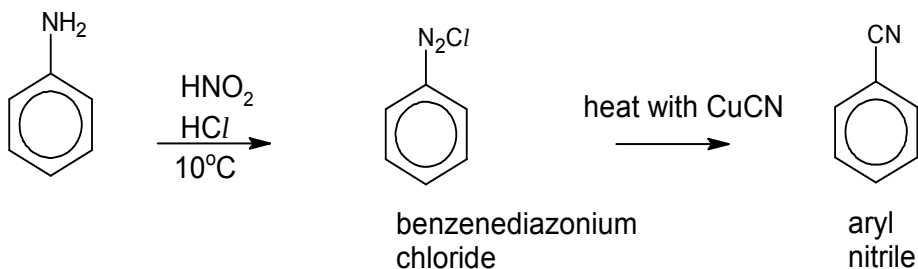


compound O

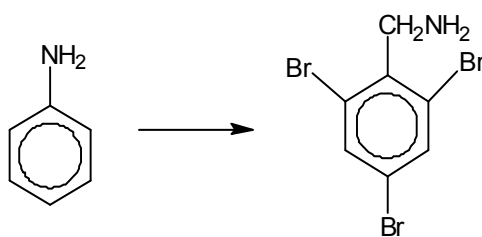
[3]

- (e) Phenylamine reacts with nitrous acid,  $\text{HNO}_2$ , and hydrochloric acid at  $10^\circ\text{C}$  to form benzenediazonium chloride,  $\text{C}_6\text{H}_5\text{N}_2\text{Cl}$ .

This is an important intermediate in the synthesis of aromatic compounds. It undergoes the Sandmeyer reaction, shown below, in the presence of a copper(I) catalyst to form aryl nitriles.



Suggest a four-step reaction sequence for the following conversion. You are to make use of the Sandmeyer reaction in one of the steps.



[4]

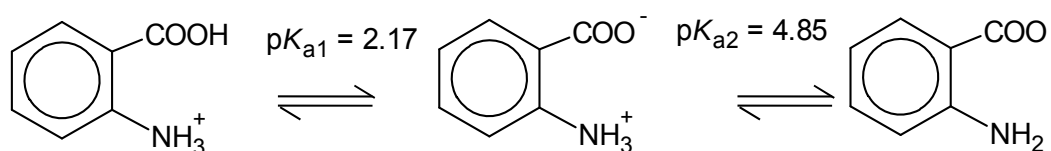
[Total:19]

## Section B

Answer **one** question from this section.

- 4 (a) Vitamin L1, also known as 2-aminobenzoic acid, can be obtained from bovine liver, and its deficiency has been associated with lactation issues in females.

The diagram below shows how the protonated form of Vitamin L1 acts as a dibasic acid, which dissociates in two stages:



Protonated form  
of Vitamin L1

- (i) Calculate the pH of 10 cm<sup>3</sup> of 0.100 mol dm<sup>-3</sup> of the protonated form of Vitamin L1. Ignore the effect of  $pK_{a2}$  on the pH of the sample. [1]
- (ii) A sample containing 10 cm<sup>3</sup> of 0.100 mol dm<sup>-3</sup> of the protonated form of Vitamin L1 was titrated with 30 cm<sup>3</sup> of 0.100 mol dm<sup>-3</sup> of NaOH.

Sketch clearly the titration curve obtained, which should include the following key points:

- initial pH;
- points where  $pK_{a1}$  and  $pK_{a2}$  occur;
- isoelectric point of Vitamin L1 and its estimated value, given that:  
isoelectric point =  $(pK_{a1} + pK_{a2}) / 2$

[4]

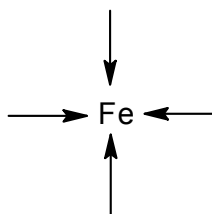
- (iii) Define what is meant by a buffer. [1]
- (iv) Draw the structures of the two chemical species present in a solution of Vitamin L1 at pH 5.00.

Hence, show by means of balanced equations, how this solution regulates the acidity on addition of H<sup>+</sup> ions and OH<sup>-</sup> ions. [3]

- (b) Iron ions can form transition metal complexes with the basic form of the amino acid, alanine. The formula of the complex formed is  $\text{Fe}(\text{CH}_3\text{CH}(\text{NH}_2)\text{COO})_2$ .

(i) State the oxidation state of Fe in the transition complex. [1]

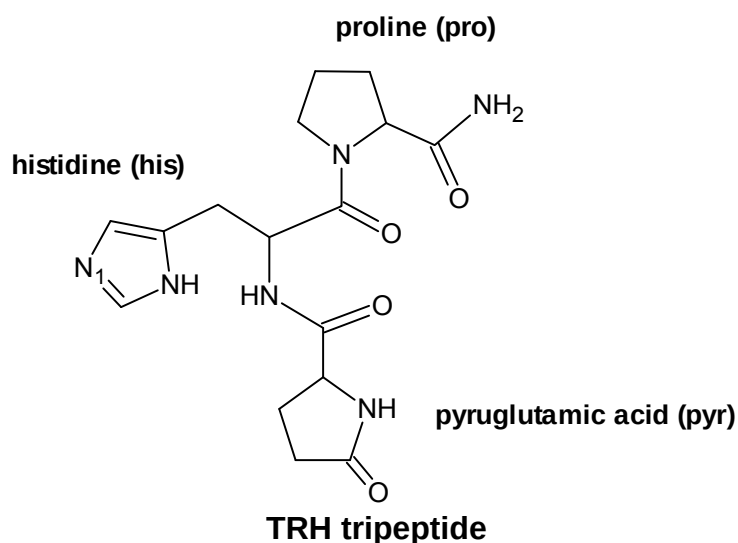
(ii) The diagram below shows the incomplete structure of the complex. Copy the diagram and draw the complete structure of this complex.



[1]

- (c) Thyrotropin-releasing hormone (TRH) is a hormone that is produced by the hypothalamus in the brain. TRH has been used clinically for the treatment of the disturbance of consciousness in humans.

TRH is a tripeptide that is made of three amino acids proline, histidine and pyruglutamic acid, as shown in the diagram below.



In the questions below, assume that the side chain for histidine (his) is inert.

(i) State the reagents and conditions for TRH to undergo hydrolysis. [1]

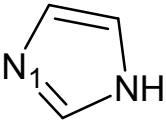
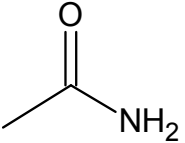
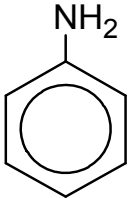
(ii) Draw the structures of the amino acids formed when TRH undergoes hydrolysis using the reagents and conditions stated in (c)(i). [3]

(iii) Explain why amino acids exist as solids with high melting points. [1]

(iv) State the hybridisation state of nitrogen atom labeled  $\text{N}_1$  in the histidine side chain of TRH. [1]

- (v) The amino acid histidine has a cyclic imidazole side-chain.

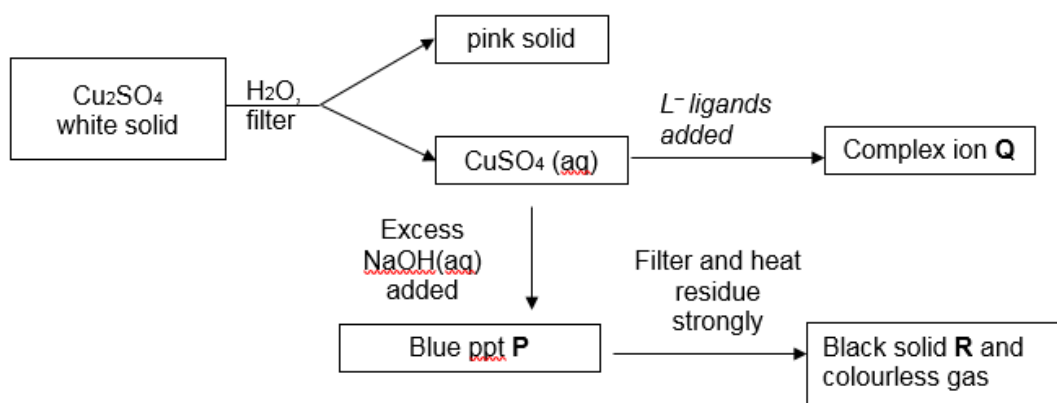
The table below compares the  $pK_b$  value of  $N_1$  nitrogen atom of imidazole ring, with the  $pK_b$  values of ethanamide and phenylamine.

| Compound name | Structure                                                                         | $pK_b$                     |
|---------------|-----------------------------------------------------------------------------------|----------------------------|
| imidazole     |  | 8.8<br>( $pK_b$ of $N_1$ ) |
| ethanamide    |  | 14.5                       |
| phenylamine   |  | 9.4                        |

Using the data provided, list the 3 compounds in increasing basicity, and suggest explanations for their relative basicities. [3]

[Total: 20]

- 5 (a) The scheme below shows the reactions of different compounds of copper.

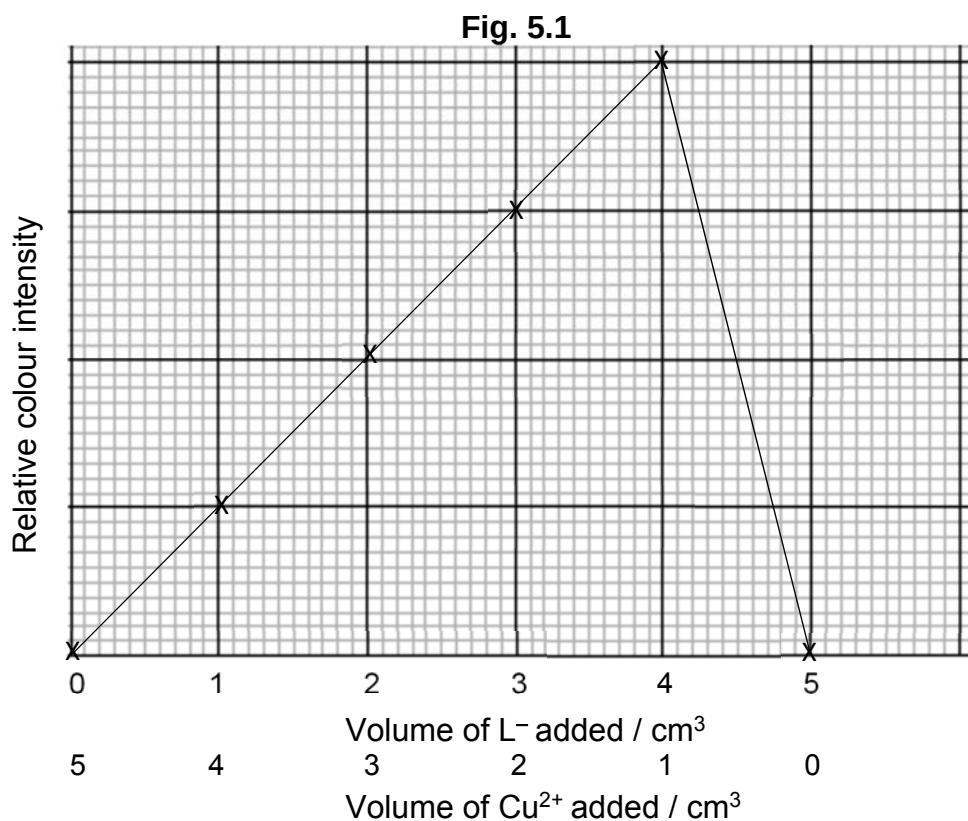


- (a) (i) Suggest the formulae of blue ppt **P** and black solid **R**. [2]

- (ii) Write an equation to show how the pink solid and  $\text{CuSO}_4$  are obtained when  $\text{Cu}_2\text{SO}_4$  is added to water. [1]

To determine the stoichiometry of the green complex ion **Q** formed between aqueous copper (II) sulfate and  $\text{L}^-$  ligands, the colour intensities of solutions containing different proportions of the ligand,  $\text{L}^-$ , and  $\text{Cu}^{2+}$  ion were measured.

The intensity of the green colour is directly proportional to the concentration of the complex ion. The higher the concentration of the complex, the higher is the colour intensity. In one such experiment, the results obtained for different volume mixtures of  $1.0 \times 10^{-3} \text{ mol dm}^{-3} \text{ Cu}^{2+}(\text{aq})$  and  $1.0 \times 10^{-3} \text{ mol dm}^{-3} \text{ L}^-(\text{aq})$  were plotted below.



(iii) State the type of reaction that occurs when complex ion **Q** is formed from aqueous  $\text{CuSO}_4$ . [1]

(iv) Based on the plotted graph in **Fig. 5.1**, determine the mole ratio of  $\text{Cu}^{2+}(\text{aq})$  to  $\text{L}^{-}(\text{aq})$  in the complex **Q**.

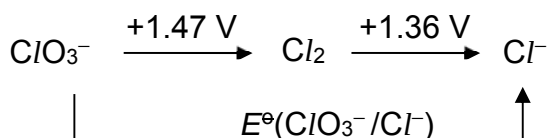
Given that complex **Q** has an octahedral shape and it retains some water molecules as ligands, suggest a formula for the complex **Q**, including its overall charge. [2]

(b) Explain why complexes of  $\text{Cu}(\text{II})$  are coloured whereas complexes of  $\text{Cu}(\text{I})$  are colourless. [4]

(c) Chlorine can exist in different oxidation states.

| Half-equation                                                                                              | $E^{\circ} / \text{V}$ |
|------------------------------------------------------------------------------------------------------------|------------------------|
| $2\text{ClO}_3^{-} + 12\text{H}^{+} + 10\text{e}^{-} \rightleftharpoons \text{Cl}_2 + 6\text{H}_2\text{O}$ | +1.47                  |
| $\text{Cl}_2 + 2\text{e}^{-} \rightleftharpoons 2\text{Cl}^{-}$                                            | +1.36                  |

A Latimer diagram shown below, summarises the standard electrode potential,  $E^{\circ}$  of different chlorine-containing species in acidic medium. The most highly oxidised form of the element is on the left, with successively lower oxidation states to the right. The electrode potentials are shown on the arrows.



The standard electrode potential of converting  $\text{ClO}_3^{-}$  to  $\text{Cl}^{-}$ ,  $E^{\circ}(\text{ClO}_3^{-} / \text{Cl}^{-})$  is **NOT** the summation of +1.47 V and +1.36 V.

Determine the number of electrons involved in the conversion of 1 mole of  $\text{ClO}_3^{-}$  to 1 mole of  $\text{Cl}^{-}$ .

Hence, given that  $\Delta G^{\circ}$  for the conversion of  $\text{ClO}_3^{-}$  to  $\text{Cl}^{-}$  is  $-150 \text{ kJ mol}^{-1}$ , calculate  $E^{\circ}(\text{ClO}_3^{-} / \text{Cl}^{-})$ . [2]

(d) *Use of the Data Booklet is relevant to this question.*

**X**,  $\text{C}_5\text{H}_6\text{O}_3$ , is soluble in dilute  $\text{NaOH}$ . **X** gives an orange precipitate with 2,4-dinitrophenylhydrazine reagent, and a yellow precipitate with warm alkaline aqueous iodine. **X** decolourises aqueous bromine.

When it is reacted with a methanolic solution of  $\text{NaBH}_4$  it gives **Y**,  $\text{C}_5\text{H}_8\text{O}_3$ . Catalytic hydrogenation of **X** gives **Z**,  $\text{C}_5\text{H}_{10}\text{O}_3$ . When **Z** is reacted with alkaline aqueous iodine, it gives  $^-\text{O}_2\text{C}(\text{CH}_2)_2\text{CO}_2^-$ .

Oxidation of 1 mole of **X** with hot acidified potassium manganate(VII) gives 1 mole of  $\text{CH}_3\text{COCO}_2\text{H}$  and 2 moles of  $\text{CO}_2$  gas.

*Additional optional information: X, Y and Z have a proton ( $^1\text{H}$ ) chemical shift value of 13.0 ppm. (Refer to Data Booklet Section 6: Typical proton ( $^1\text{H}$ ) chemical shift values ( $\delta$ ))*

Suggest structures for **X**, **Y** and **Z**. Explain your reasoning clearly.

[8]  
[Total: 20]

**End of paper**